The Puppy Puzzle

Focus question: How can puppies from the same parents be alike, but not identical?

Notice and wonder

Model: Show your puppy photos (or the data on the next slide). Two notices and two wonders each; take 3 aloud. Steer toward: alike in some traits, different in others.

Compare the traits

Unit 2.2 · Day 1 · Glue in · write on it · KEY Puppy litter data

	Fur color	Ear shape	Pattern
Mom	Black	Floppy	Spotted
Dad	Brown	Pointed	Solid
Puppy A	Black (M)	Floppy (M)	Solid (D)
Puppy B	Brown	Pointed	Spotted
Puppy C	Black	Pointed	Spotted
Puppy D	Brown	Floppy	Solid

Every puppy has some traits like Mom and some like **Dad**.

No two puppies have the same **combination** of traits.

The Puppy Puzzle (continued)

My first model

Model: A model of how traits get from parents to puppies. Copy the problem: The puppies share some traits, but they are not identical. How can offspring inherit traits from their parents and still be different from one another? Draw: Parent 1, Parent 2, and two offspring. Use arrows to show information moving from the parents to the offspring. Label one shared trait and one different trait. It's OK to guess. We fix this model on Day 10.

Three possible explanations

Model: Three small sketch boxes. 1. Information gets mixed: offspring may get different combinations of information from their parents. 2. Information changes: it may sometimes change when it is copied. 3. The environment has an effect: sunlight, temperature, or food may affect how an animal looks.

Key idea

Parents → **information** → Offspring → similarities and **differences**

Exit ticket

One explanation for why siblings can be different is **the information is combined in different ways**.

The evidence from the puppies is **the information changes**.

Mixing Information

Focus question: If offspring get information from the same two parents, why might each get a different combination?

Warm-up

Idea 1: information is **combined** in different ways. Idea 2: information **changes**. Idea 3: the **environment** has an effect.

Make possible offspring

Unit 2.2 · Day 2 · Glue in · write on it · KEY Card table: 3 offspring

Draw one card from each of the six piles. Write what you drew, then put the cards back. Offspring 1 is done for you.

Offspring	Fur color (P1 + P2)	Ear shape (P1 + P2)	Pattern (P1 + P2)
1 (done for you)	Black + White	Floppy + Pointed	Spotted + Solid
2			
3			

Mixing Information (continued)

Analyze the pattern

Every offspring got information from **both** parents.

Each offspring got a **different** combination of cards.

Different combinations can lead to different **traits**.

Revise our model

Model: Revise your Day 1 model. Show: Parent 1, Parent 2, information from each parent, the information being combined, two possible offspring, one similarity, and one difference.

Three sources of variation

1. New **combinations** of alleles. 2. **errors** when DNA is copied. 3. Mutations caused by the **environment**.

Genetics Notes, Part 1: Genes and Alleles

Focus question: What are genes and alleles, and what do dominant and recessive mean?

Warm-up

A **chromosome** is one long DNA molecule. A **gene** is a section of DNA that codes for one protein, and proteins help produce a **trait**.

Heredity

Genetics is the study of **heredity**. Heredity is the passing of **traits** from **parents** to **offspring**. The width of a tiger's stripes is a trait **inherited** from its parents.

Chromosomes

Inside each nucleus of a cell are thread-like structures called **chromosomes**. Chromosomes come in **pairs**. You usually have **two** copies of each chromosome, one from each parent.

Genes make proteins make traits

Genes found within a tiger's **DNA** produce **proteins** that determine the width of a tiger's **stripes**.

Genetics Notes, Part 1: Genes and Alleles (continued)

Alleles

Model: Guided notes and a chromosome sketch. In your notebook, color each set of alleles on the chromosome pair a different color.

Individuals inherit **two** forms of each **gene**, one from each parent. The different variations of a gene are known as **alleles**. Alleles are represented with lowercase and uppercase **letters**.

Dominant vs. recessive

Dominant: **always** expressed when present. Shown as **capital** letters.

Recessive: expressed if **two** copies are present. Shown as **lowercase** letters.

Black or brown cats

Model: Color the cat heads: BB, Bb, bb. A black coat (B) is dominant to a brown coat (b) in cats. A cat needs only one B to be black. A cat needs two b alleles to be brown. In your notebook, color each cat head to show its color.

A cat with Bb is **black** because **B** is dominant. A cat needs **two** copies of b to be brown.

Exit ticket

An allele that shows up when only one copy is present is **dominant**. A recessive trait shows only when **two** copies are present.

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares

Focus question: How can a Punnett square show the possible allele combinations?

Warm-up

A dominant allele is written with a **capital** letter. A recessive allele is written with a **lowercase** letter.

Genotype and phenotype

Tt is **tall** because T is dominant. tt is **short**.

Model: Table: genotype to phenotype. In pea plants, T (tall) is dominant to t (short). TT is done for you.

Unit 2.2 · Day 4 · Glue in · write on it · KEY

Genotype to phenotype: pea plants

Genotype	Phenotype
TT	Tall (done for you)
Tt	**Tall**
tt	**Short**

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares (continued)

Homozygous and heterozygous

Model: Guided notes and six genotypes to color. Genotypes can be homozygous or heterozygous. In your notebook, color the homozygous genotypes green and the heterozygous genotypes orange.

Punnett squares

Unit 2.2 · Day 4 · Glue in · copy of the board · KEY

Punnett squares: steps and worked example

1. Write one parent's two alleles across the top. **2.** Write the other parent's two alleles down the side. **3.** Fill each box with the allele from its column and the allele from its row.

Worked example: Tt × Tt

	T	**t**
T	TT	Tt
t	Tt	tt

Result: TT, Tt, Tt, tt. That is **3 tall to 1 short**.

Each box in a Punnett square shows one possible ____ of alleles.

Practice

Cross 1: 2 of the 4 boxes are tall and 4 are short.

Cross 2: _____ of the 4 boxes are tall.

Exit ticket

In a Tt × Tt cross, 1 out of 4 boxes show a short plant (tt).

Monster Genetics, Part 1: Flip a Coin, Build a Monster

Focus question: How can two coin flips build a monster's genotype?

Warm-up

Each sex cell gets **one** allele. A coin flip can choose between B and **b**.

How the lab works

Unit 2.2 · Day 5 · Glue in · reference and write on it · KEY Monster Genetics: the coin rule and Table 2

Trait	Allele 1 (heads)	Allele 2 (tails)
Eyes	Two small eyes (E)	One large eye (e)
Eye color	Red (R)	White (r)
Skin color	Green (G)	Blue (g)

Trait	Genotype	Phenotype
Eyes	ee	One large eye (done for you)
Eye color	**rr**	White
Skin color	Gg	**Green**
Tail shape	**cc**	Straight
Tail color	Pp	**Purple**

Monster Genetics, Part 1: Flip a Coin, Build a Monster (continued)

Build the female monster

Use Table 1 on your Monster Genetics sheet.

Model: Sheet: "Corrected_Monster Genetics (Mendellian genetics only)" (the Nov 16 planner link). Circulate for dominant and recessive mix-ups.

Fill in the male monster

Use Table 2 (glued in above) and your Monster Genetics sheet.

Model: Key (all 12): ee one large eye; rr white; Gg green; cc straight; Pp purple; TT has a tail; ss round teeth; Ff four toes; ww white horn; YY pointy ears; nn two ears; Ll long claws.

Compare

Our monsters are different because the flips gave different **combinations** of alleles.

Exit ticket

Each coin flip stood for the **allele** a sex cell passes on. Two flips gave the monster's **genotype**.

Monster Genetics, Part 2: Predict the Baby

Focus question: How do sex cells and fertilization mix alleles, and how can we predict the baby?

Warm-up

A Gg parent makes sex cells with **G** or **g**.

Sex cells and fertilization

Meiosis makes **sex cells** with one allele of each gene. Crossing over happens during meiosis, when paired chromosomes **swap** pieces of DNA. Fertilization joins two sex cells to make a new **combination** of alleles.

Punnett squares for the baby

Unit 2.2 · Day 6 · Glue in · write on it · KEY Punnett square: skin color and eyes

	G	**g**
G	GG	Gg
g	Gg	gg

Eyes: ee × ee

	e	**e**
e	ee	ee
e	ee	ee

Monster Genetics, Part 2: Predict the Baby (continued)

Percent chance

In a Gg × Gg cross, **75**% of the boxes are green and **25**% are blue.

Class data

Unit 2.2 · Day 6 · Glue in · reference · KEY Prediction vs. class count

Baby's skin color	Punnett prediction	Class count (sample)	Class percent
Green	75%	22 of 30	73%
Blue	25%	8 of 30	27%

The Punnett square predicted **75**% green. The class got about **73**% green. The numbers are close but not exact because every flip is **random**.

When Copying Goes Wrong

Focus question: How do copying errors and UV light change genes, and when are the changes inherited?

Warm-up

A mutation is a **change** in the DNA sequence. A mutation can change an **amino acid** in a protein.

How do mutations happen?

A **copying** error that is not fixed becomes a mutation. UV light and some **chemicals** can also damage DNA.

Four kinds of mutations

Frameshift: the **mRNA** codons shift forwards or **backwards**, changing many amino acids.

Nonsense: a **stop** codon **terminates** translation early, often leaving a non-functional protein.

When Copying Goes Wrong (continued)

UV and mutations

Unit 2.2 · Day 7 · Glue in · reference · KEY UV light and mutations

Hours of UV light	Mutations per 1,000 skin cells
0 (none)	2
1	9
2	21
4	40

Who passes it on?

Unit 2.2 · Day 7 · Glue in · write on it · KEY Who passes it on? Five cases

Case	Where the DNA changed	Inherited?
A	UV hits a skin cell	No (done)
B	Copy error in a sperm-making cell	**Yes**
C	Copy error in a muscle cell	**No**
D	X-rays hit an egg-making cell	**Yes**
E	Smoke chemical hits a lung cell	**No**

Explain it

UV light can change a gene by ***damaging*** its DNA.

The change is inherited only if it happens in a cell that makes ***eggs or sperm***.

Where Did This Variation Come From?

Focus question: How can data show where a genetic variation came from?

Clues for each source

New combination: every allele in the offspring was **already** in a parent.

Copying error: a **new** allele that neither parent has, and no extra exposure.

Environmental mutation: **more** mutations in the group that was exposed.

Case 1: the puppy litter

Unit 2.2 · Day 8 · Glue in · reference · KEY Case data: three cases

Case 1: puppy litter

	Alleles for fur color (B = black, b = brown)
Mom	B and b
Dad	B and b
Puppies	4 black (BB or Bb), 2 brown (bb)

Case 2: DNA of a puppy and its parents

	Allele 1	Allele 2
Mom	CAT	CAT
Dad	CAT	CGT
Puppy	CAT	CCT

Case 3: X-rays and mice

Group of mice	X-ray dose	New mutations per 1,000 sperm cells
A	none	3
B	low	12
C	high	25

Where Did This Variation Come From? (continued)

Case 2: a new base

Use "Case data: three cases" (glued in above).

The puppy's CCT allele is **new** because neither parent has it. The most likely source is a **mutation**, not a new combination.

Case 3: X-rays and mice

Use "Case data: three cases" (glued in above).

The group with a high X-ray dose had **25** mutations. The group with no X-rays had **3**.

The extra mutations came from **the X-rays**.

Write your claim

Claim: The extra mutations came from **an environmental mutation**.

Evidence: The high-dose group had **25** mutations per 1,000 sperm cells, but the group with no X-rays had **3**.

Reasoning: If copying errors alone caused every mutation, the groups would have **similar** numbers.

Back to the Puppies and Make-Up Work

Focus question: How do I defend my claim against another explanation?

How to defend a claim

A counterclaim is **another** explanation. To defend my claim I show why the **counterclaim** does not fit the data.

The new coat pattern

Unit 2.2 · Day 9 · Glue in · reference · KEY

The new coat pattern: two kennels

Kennel	Puppies checked	Puppies with a new coat pattern (in neither parent)
A (chemical in the water)	1,000	14
B (clean water)	1,000	3

Evidence: Kennel A had **14** new patterns per 1,000 puppies. Kennel B had **3**.

Defense: Copying errors happen in **both** kennels, so they cannot explain the **11** extra patterns.

Back to the Puppies and Make-Up Work (continued)

Fix my Day 1 model

Model: Redraw with all three sources. Redraw your Day 1 model. Check off each part as you add it: ☐ new combinations (sex cells and fertilization) ☐ copying errors ☐ mutations from the environment ☐ a note that only changes in eggs or sperm are inherited ☐ the puppy traits

Write It: the puppy puzzle

Claim: Puppies from the same parents are alike but not identical because of **new combinations of alleles**.

Evidence: In the card activity, each offspring got a different **combination** of alleles.

Reasoning: Sex cells and fertilization make new combinations. Copying errors and **environmental mutations** can also change DNA, and a change is inherited only if it is in **eggs or sperm**.

Make up missing work

Model: Short periods today (37 minutes). The 15-minute phenomenon wrap-up (LS3-2.5) comes first; the rest of the period is for catching up on missing work. Reviews are saved for Monday, Nov 30.

Review Day: Prepare for the CSA

Focus question: Which CFA reviews will best get me ready for the CSA?

Check my CFAs

Unit 2.2 · Day 10 · Glue in · write on it · KEY

My target tracker

Write your CFA score out of 5. Below a B (fewer than 4 of 5)? Write Yes under Review. At the end of class, rate each target 1 to 4 (4 = I could teach it).

Target	CFA score (of 5)	Review?	Rating (1 to 4)
LS3-2.1 three sources			
LS3-2.2 sex cells and fertilization			
LS3-2.3 copying errors and UV			
LS3-2.4 read data, make a claim			

My reviews

Model: This whole 47-minute class is for the reviews from low-scoring CFAs, to prepare for the CSA tomorrow. Pull small groups by lowest target. Students with every review done go to the enrichment.

Review Day: Prepare for the CSA (continued)

Enrichment: environment and traits

Unit 2.2 · Day 10 · Glue in · reference · KEY Hydrangea soil data

Soil pH	Flower color (same DNA)
5.0	Blue
6.0	Purple
7.0	Pink

The plants have the same DNA. Flower color varied because the soil changed how the gene works.

Self-check

Rate each target 1 to 4 on your tracker (glued in above).